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import pandas as pd
import pickle

from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import Ridge
from sklearn.metrics import r2_score
from google.colab import files

# Upload dataset during runtime
uploaded = files.upload()

file_name = list(uploaded.keys())[0]
df = pd.read_csv(file_name)

# Region encoding for feature consistency
if "Region" in df.columns and df["Region"].dtype == "object":
    df["Region"] = df["Region"].map({"Sidi-Bel Abbes": 0, "Bejaia": 1})

# Feature selection based on correlation with FWI
corr = df.corr(numeric_only=True)
print(corr["FWI"].sort_values(ascending=False))

X = df.drop("FWI", axis=1)
y = df["FWI"]

# Train-test split for generalizability
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# StandardScaler for normalization
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Save scaler for deployment consistency
with open("scaler.pkl", "wb") as f:
    pickle.dump(scaler, f)

# Ridge Regression to handle multicollinearity
ridge = Ridge()

# Alpha tuning for better validation performance
param_grid = {"alpha": [0.01, 0.1, 1, 10, 100]}
grid_search = GridSearchCV(ridge, param_grid, cv=5, scoring="r2")
grid_search.fit(X_train_scaled, y_train)

best_model = grid_search.best_estimator_
y_pred = best_model.predict(X_test_scaled)

print("Best Alpha:", grid_search.best_params_)
print("Validation R2 Score:", r2_score(y_test, y_pred))

# Save trained model for deployment
with open("ridge.pkl", "wb") as f:
    pickle.dump(best_model, f)

print("scaler.pkl and ridge.pkl saved successfully")
```

cleaned_fwi...UTPUT.csv

cleaned_fwi_dataset OUTPUT.csv(text/csv) - 12869 bytes, last modified: 3/2/2026 - 100% done

Saving cleaned_fwi_dataset OUTPUT.csv to cleaned_fwi_dataset OUTPUT.csv

FWI 1.000000

ISI 0.922895

DMC 0.875864

BUI 0.857973

DC 0.739521

FFMC 0.691132

Temperature 0.566670

day 0.350781

Region 0.197102

month 0.082639

Ws 0.032368

Rain -0.324422

RH -0.580957

Classes -0.712106

year NaN

Name: FWI, dtype: float64

Best Alpha: {'alpha': 1}

Validation R2 Score: 0.9834103862506142

scaler.pkl and ridge.pkl saved successfully